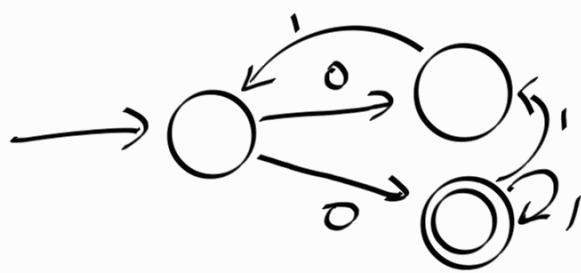


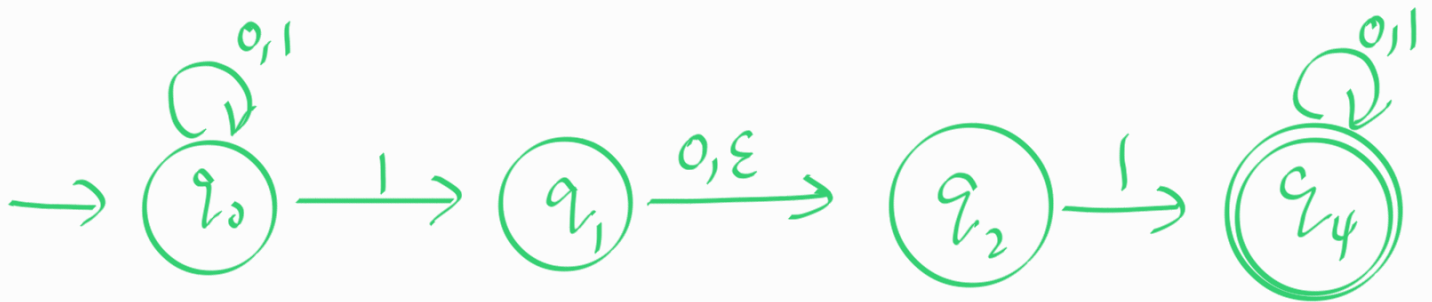
Deterministic: Given the current state and the input symbol, we know exactly which state to go to:



$\Sigma = \{0, 1\}$

Not a DFA

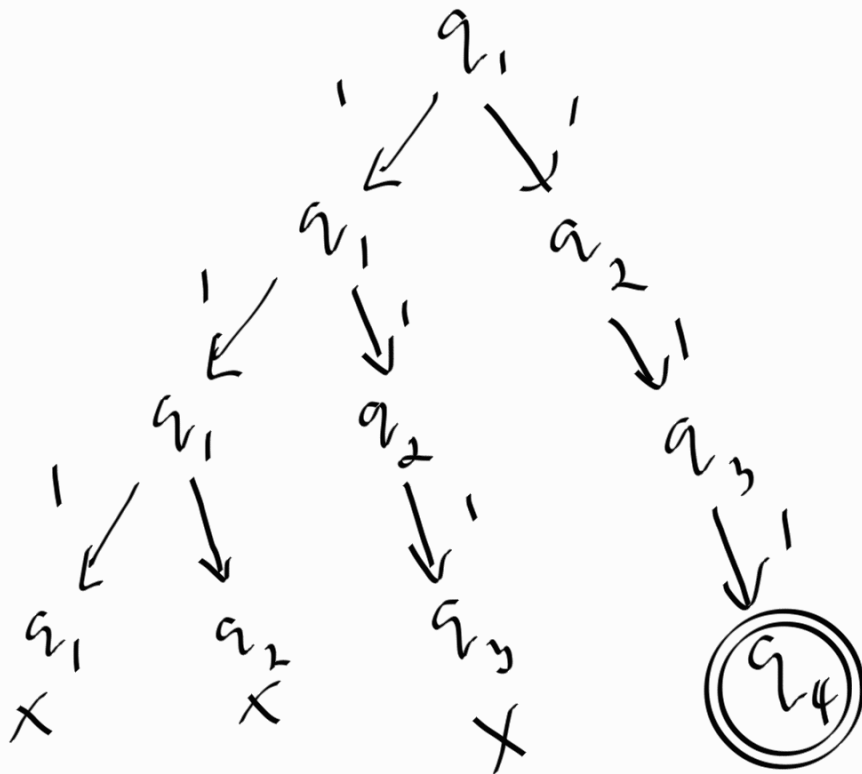
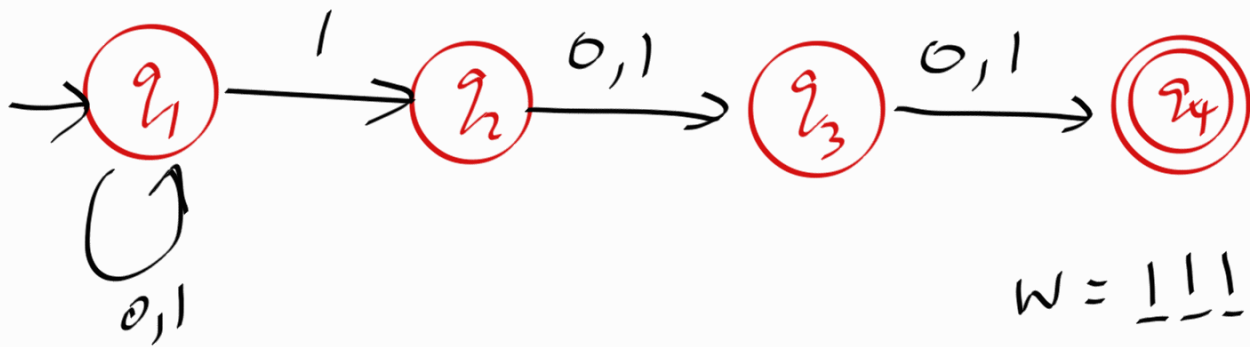
Non deterministic



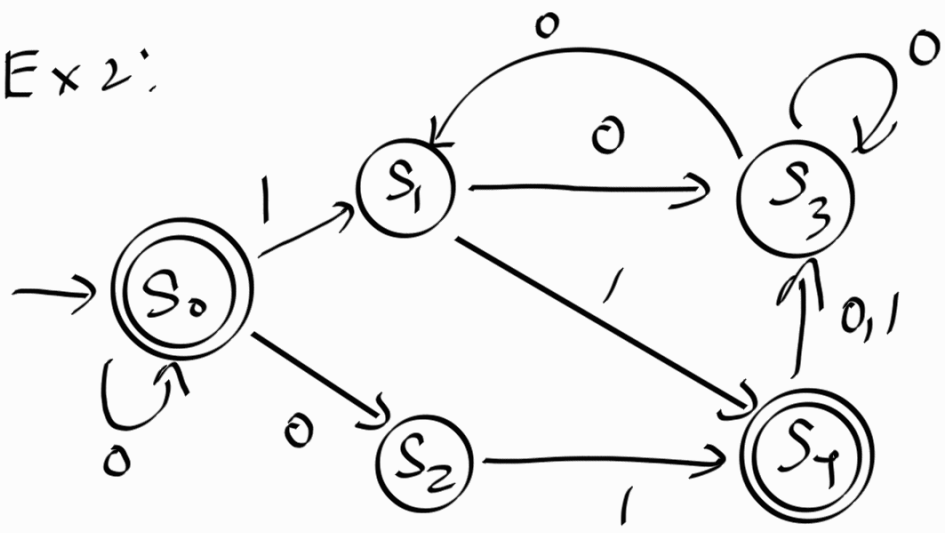
1. Many arrow from a state on the same symbol.
2. Transitions on empty string

How does NFA compute?

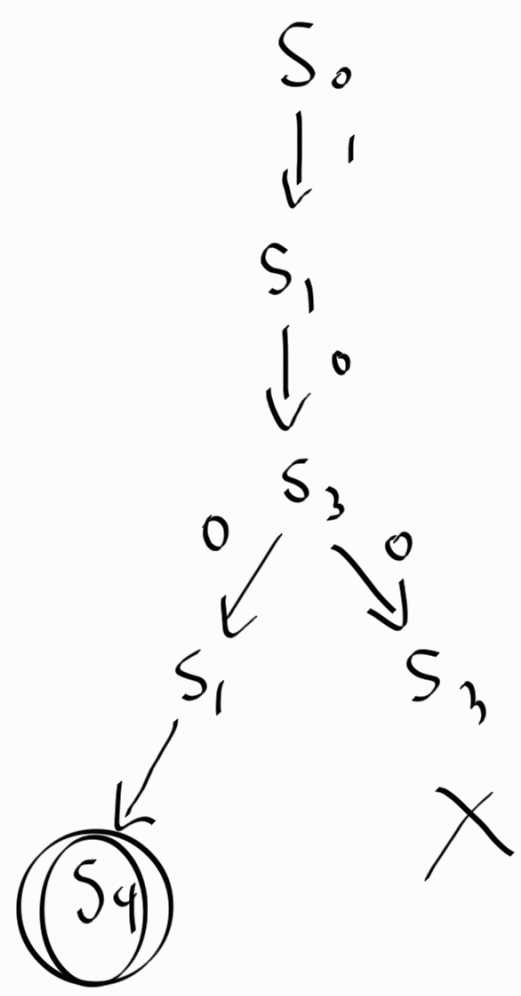
- If any of the accept states can be reached by reading the input x in any way, we say NFA accepts x .



Ex 2:

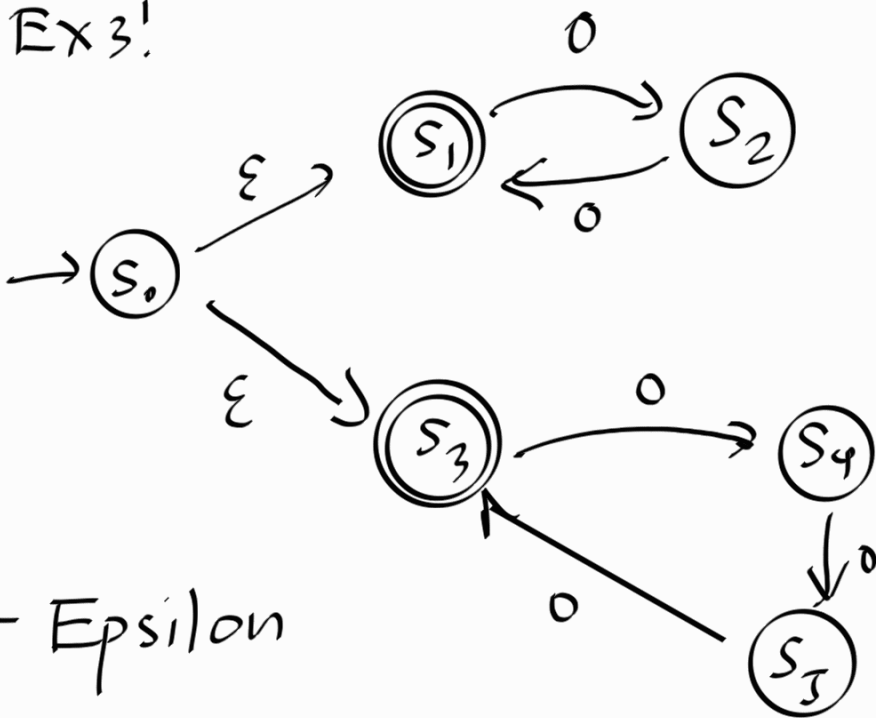


$w = \underline{1} \underline{0} \underline{0} \underline{1} \checkmark$

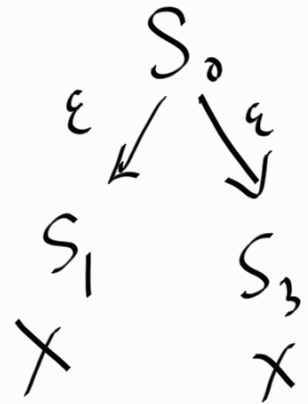


Ex 3!

$$\Sigma = \{0, 1\}$$



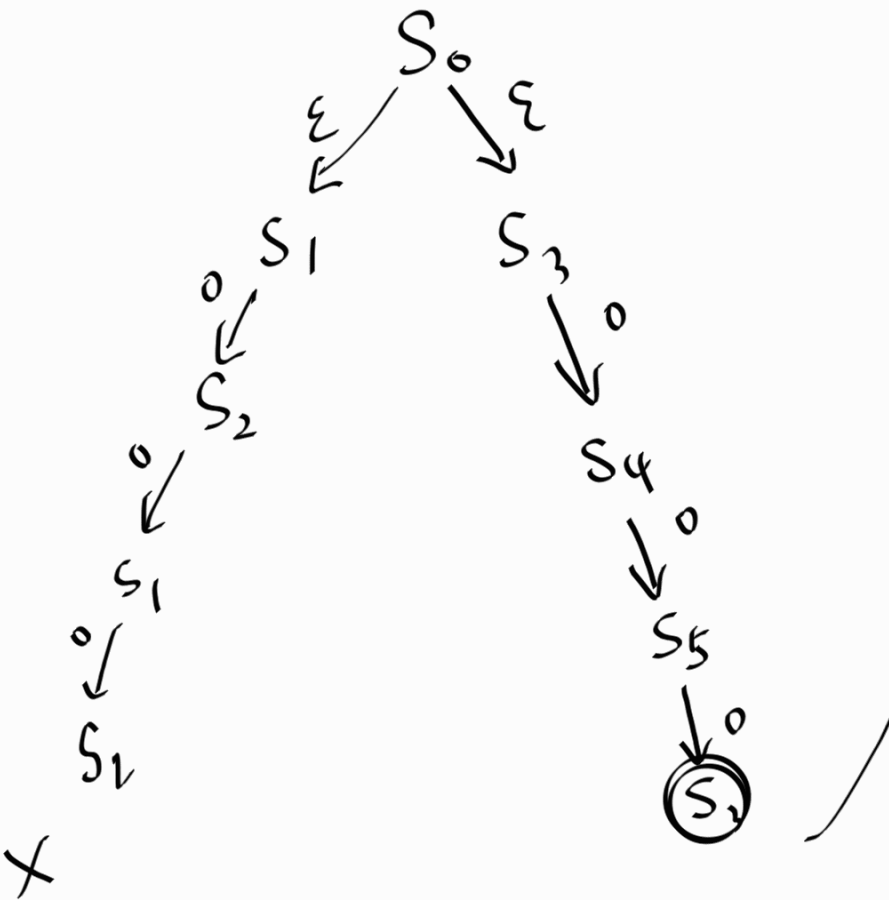
$$w = \underline{1}001$$



ϵ -Epsilon

$$\epsilon = ""$$

$$w = \underline{0}\underline{0}\underline{0} \quad \checkmark$$



Definition of NFA.

NFA is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$
s.t

1. Q is the set of states
2. Σ is the finite alphabet.
3. $\delta: Q \times \Sigma_\epsilon \longrightarrow P(Q)$

$$\Sigma_\epsilon = \Sigma \cup \{\epsilon\}$$

4. $q_0 \in Q$ is the start state
5. $F \subseteq Q$ is the set of accepting states

How does a NFA accept a string?

Let $N = (Q, \Sigma, \delta, q_0, F)$ be an NFA,
and $w = \langle w_1, w_2, w_3, \dots, w_n \rangle$ be a string,

$$\forall i \in [n]: w_i \in \Sigma$$

N accepts w if,

$\exists r_0, r_1, r_2, \dots, r_n \in Q$ st

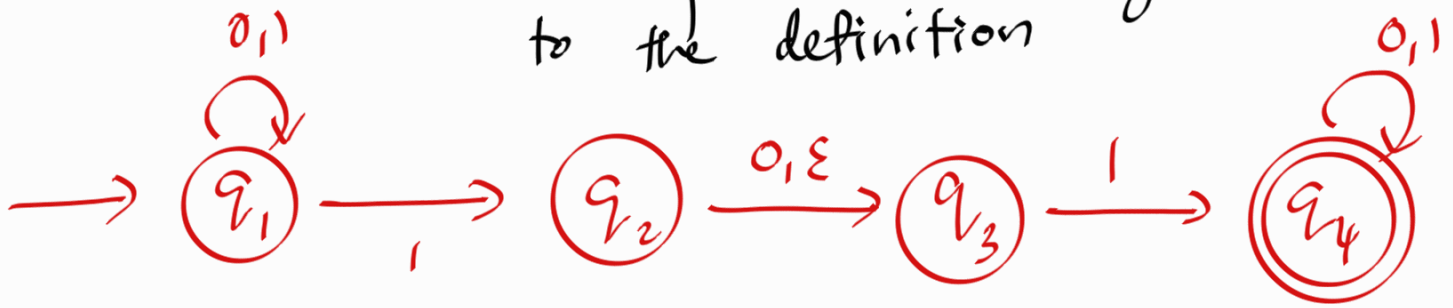
1. $r_0 = q_0$

2. $r_i \in \delta(r_{i-1}, w_i) \quad \forall i \in [1 \dots n]$

3. $r_n \in F$



Let's map the following NFA to the definition



1. $Q = \{q_1, q_2, q_3, q_4\}$

2. $\Sigma = \{0, 1\}$, $\Sigma_\epsilon = \{0, 1, \epsilon\}$

3. $\delta: Q \times \Sigma_\epsilon \rightarrow \underline{P(Q)}$

	ϵ	0	1
q_1	$\emptyset$	$\{q_1\}$	$\{q_1, q_2\}$
q_2	$\{q_3\}$	$\{q_3\}$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_4\}$
q_4	$\emptyset$	$\{q_4\}$	$\{q_4\}$

4. $q_1 \in Q$ is the start state

5. $F = \{q_4\}$

Let's do some examples

1. Design a DFA that accepts the following language.

$$C = \{ w \mid w \text{ has an odd \# of } a\text{'s and ends with } b \}$$

$$\Sigma = \{ a, b \}$$

